

Continuous Solvent/Detergent Virus Inactivation Using a Packed-Bed Reactor

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Continuous virus inactivation (VI) remains one of the missing pieces while the biopharma industry moves toward continuous manufacturing. The challenges of adapting VI to the continuous operation are two-fold: 1) achieving fluid homogeneity and 2) a narrow residence time distribution (RTD) for fluid incubation. To address these challenges, a dynamic active in-line mixer and a packed-bed continuous virus inactivation reactor (CVIR) are implemented, which act as a narrow RTD incubation chamber. The developed concept is applied using solvent/detergent (S/D) treatment for inactivation of two commonly used model viruses. The in-line mixer is characterized and enables mixing of the viscous S/D chemicals to $\pm 1.0\%$ of the target concentration in a small dead volume. The reactor's RTD is characterized and additional control experiments confirm that the VI is due to the S/D action and not induced by system components. The CVIR setup achieves steady state rapidly before two reactor volumes and the logarithmic reduction values of the continuous inactivation process are identical to those obtained by the traditional batch operation. The packed-bed reactor for continuous VI unites fully continuous processing with very low-pressure drop and scalability.

processing due to economic benefits,^[1–3] the ability to control the quality of the product,^[4,5] and to automate processes. A compromise must be found between sufficient time for VI and product loss; therefore, reactors with narrow residence time distribution (RTD) must be used to avoid unnecessary long incubations during continuous operations. While several technologies have been developed for continuous operation in the upstream^[6,7] and downstream processing,^[8,9] virus clearance operations have been neglected. At the same time, continuous VI has also been highlighted as the missing piece of the continuous processing puzzle.^[10–12]

VI can be accomplished through different methods, such as solvent/detergent (S/D) treatment.^[13] S/D treatment has been used for decades in the blood product industry^[14] and is a superior method, particularly for acid-labile recombinant products though only effective against enveloped viruses. VI using an

S/D mixture was first demonstrated by Horowitz et al. in 1985.^[15,16] Nowadays, a mixture of tri-*n*-butyl phosphate (TnBP) and triton X-100 (TX-100) both at 1% v/v concentration has become the industry standard.^[17] However, additional detergents might be included in some cases, e.g., 0.3% v/v polysorbate 80 (PS80).^[13] In batch operation, the S/D chemicals are homogenized with the process intermediate in a primary vessel and subsequently transferred to a secondary vessel, in which the VI incubation takes place: the “two-vessel” strategy is preferred by the regulatory authorities and is the industry's current practice.^[12] In manufacturing processes, the specified VI incubation ranges from 60 min to 8 h.^[13]

The challenge of adapting a batch VI to continuous operation lies on tightly regulated process parameters to be attained. These challenges are twofold: fluid homogeneity and minimum incubation time. In the case of S/D treatment achieving homogeneity is not trivial due to the S/D chemicals' viscosity, but can be achieved through active or static in-line mixing. Incubation time, on the other hand, represents a much greater challenge. Any batch incubation time becomes a time distribution in a continuous setup. The simplest way to control the minimum incubation time is running two VI vessels in staggered mode, implemented in Pall's Cadence Virus Inactivation System.^[18] Besides, not being truly continuous, the proposed system ignores the “two-vessel”

1. Introduction

Continuous virus inactivation (VI) remains a major challenge while the biopharma industry is moving toward continuous

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strategy, which is likely to pose regulatory acceptance challenges. In 2015, Klutz et al.^[19] suggested a coiled flow inverter (CFI) reactor for VI.^[20] The CFI concept had been described before,^[21] though in a different context, and it consists of a coiled tube with alternating coiling direction. The developed secondary flow pattern (Dean vortices) enables radial mixing and thereby a narrower RTD compared with a straight tube. More recently, Orozco suggested a similar approach, Jig in a Box (JIB), for continuous VI.^[22–24] The JIB can be regarded as an adaptation and optimization of a CFI arranged in a 3D fashion. The curvature and direction changes induce a secondary flow pattern and momentum shifts, which contribute to narrower RTD. Segregation of fluid parcels inside a tubular reactor, either by immiscible fluid or mechanically, has also been proposed^[25] but execution at scale would be challenging.^[26] The RTD for any given reactor can be assessed by a pulse of non-interacting tracer and calculation of the *F*-curve, the cumulative RTD function.^[27] The ratio between the time at 50% cumulative RTD curve ($F_{0.5}$) and the time at 0.5% cumulative RTD curve ($F_{0.005}$), $t[F_{0.5}]/t[F_{0.005}]$, has been used to measure and compare the RTD narrowness,^[20,22] where $t[F_{0.5}]/t[F_{0.005}] = 1$ for an ideal plug flow. The CFI and the JIB have reported $t[F_{0.5}]/t[F_{0.005}]$ values ranging from 1.037 to 2.043. However, both these systems rely on secondary flow patterns for narrow RTD, which leads to limited scalability due to the lack of design and operation correlations, as recently highlighted for the CFI case.^[28] Furthermore, scaling up (or down) such a continuous VI process would require building a new custom-designed reactor for that specific scale. Additionally, the pressure drop might become a limiting factor at larger scales due to the long thin flow path.^[22]

Our approach is based on an incubation chamber in the form of a packed bed of non-porous beads that narrows the RTD. We previously characterized the performance of the continuous virus inactivation reactor (CVIR) in depth and the impact of bead size, column geometry, and bead material was optimized for low RTD (unpublished data). The $t[F_{0.5}]/t[F_{0.01}]$ for a small-scale 9.40 mL CVIR is 1.13 when using 300 μm non-porous beads (or 1.15 if $t[F_{0.5}]/t[F_{0.005}]$ is considered). Note that 9.40 mL is the volume for continuous VI application, while the geometrical volume of the column is 26.5 mL, which includes also the bead volume (see Section 2.2.1 for details). However, the CVIR can be easily scaled up like a chromatography column. One specific advantage of the CVIR is that scale-up improves RTD. As in liquid chromatography, longer columns lead to improved efficiency (typically more of theoretical plates and thus smaller second moment).^[29] When the CVIR was scaled up to a 610 mL reactor (geometrical volume 1.75 L), the $t[F_{0.5}]/t[F_{0.01}]$ was reduced to 1.06 using the same bead size, or 1.07 if $t[F_{0.5}]/t[F_{0.005}]$ is considered (unpublished data). The proposed CVIR is simple, easy to operate, and can be realized in a disposable manner. Like other systems, the CVIR operates in a truly continuous fashion making integration into a continuous process easier.

In the present work, we describe the implementation of a narrow RTD reactor for a continuous VI using S/D treatment. In this proof-of-concept study, two industry-relevant model viruses are used to assess the efficacy of the CVIR and compare it against the currently used batch process. Effectiveness is assessed by characterizing the VI kinetics as recommended for a validation study.^[30] To obtain meaningful kinetic data the S/D chemicals' concentrations were reduced

to 5% of those used in a manufacturing process; otherwise, full inactivation would be obtained in the first 1–5 min.^[13] A series of controls are performed to confirm that the observed VI is the result of the S/D action and not due to one of the components of the novel system. Although this work is focused on S/D treatment, the technology described here can be applied for other VI methods.

2. Experimental Section

2.1. Materials

2.1.1. Reagents

All chemicals were of analytical grade and were purchased from Merck Millipore (Germany), unless otherwise stated. For the S/D mixture pharmaceutical-grade TX-100 (1.08643.2500; Merck Millipore), pharmaceutical-grade PS80 (8.17061.9010; Merck Millipore), and TnBP (8.18604.0100; Merck Millipore) were used.

2.1.2. Test Item

The test item is a surrogate of a process intermediate from a biopharmaceutical process. The test item buffer consisted of 20 mM 2-(*N*-morpholino)ethanesulfonic acid (MES), 400 mM NaCl, 10 mM $\text{CaCl}_2 \cdot 2\text{H}_2\text{O}$, and 0.1% (*v/v*) PS80, pH 6.35 ± 0.10 . The conductivity was adjusted to $35 \pm 5 \text{ mS cm}^{-1}$. The test item consisted of $0.60 \pm 0.01 \text{ g L}^{-1}$ human serum albumin (HSA) in test item buffer. HSA was added to serve as a surrogate for protein biotherapeutics, either produced in a recombinant cell line or plasma-derived. The S/D stock mixture consisted of 0.438% (*v/v*) TX-100, 0.138% (*v/v*) PS80, and 0.134% (*v/v*) TnBP. The S/D stock mixture is diluted tenfold at the in-line mixer to the target final concentrations. It is important to note that the final concentrations used in this study during the incubation steps were 5% of those used in a manufacturing environment and commonly reported in the literature.^[13]

2.1.3. Viruses and Cells

The bovine viral diarrhea virus (BVDV) strain NADL (ATCC VR-1422) was propagated on MDBK (ATCC CCL-22) and titrated on BT cells (ATCC CRL-1390). For preparation of the VSS used for spiking experiments, the virus-containing suspension was harvested 2 days postinfection and centrifuged to remove cellular debris, before ultracentrifugation at $360\,000 \times g$ and 4°C for 1 h. VSS was resuspended in Tris/HCl. For titration, Dulbecco's modified Eagle's medium (DMEM; Gibco, USA) with 4.5 g L^{-1} D-glucose, 10% horse serum, 0.5 mM, L-glutamine, 100 mg L^{-1} gentamycin sulfate, 2% sodium bicarbonate, 1% nonessential amino acids (NEAA), and 1 mM sodium pyruvate was used. Xenotropic murine leukemia virus (X-MuLV) was propagated on *Mus dunni* cells (ATCC CRL-2017), which were maintained in DMEM with 10% fetal calf serum (PAA, Austria), 1% L-glutamine, 1% sodium pyruvate, 1% NEAA, and 2% sodium bicarbonate. All media contained 1% gentamicin. After incubation of the infected cell lines in CO_2 - and humidity-controlled incubators, virus stocks were harvested when more than 70% of

cells showed a cytopathic effect by a collection of the medium supernatant. The virus particles in the supernatant were pelleted by ultracentrifugation ($66\,000\times g$ at 17°C for 90 min) and resuspended in $1\times$ Hank's balanced salt solution buffer (Gibco). X-MuLV was titrated on PG4 cells (ATCC CRL-2032). Both selected viruses (X-MuLV and BVDV) are suggested as model viruses in regulatory guidelines^[30,31] and, therefore, reflect general industry practices. Additionally, rodent-derived cell lines are the most commonly used cell lines for the manufacture of biotechnology products. Such cell lines might contain endogenous retrovirus particles or retrovirus-like particles. Therefore, the capacity of the manufacturing process to remove and/or inactivate such particles must be demonstrated accordingly with the guidelines in place. For this reason, X-MuLV is considered as a specific model^[30,31] and often used. BVDV is a known contaminant of fetal bovine serum,^[32] which might have been used in cell line development or cell production. Therefore, it is a requirement to demonstrate effective virus clearance to guard against potential bovine contaminants.

2.2. Methods

2.2.1. CVIR Characterization and Controls

The CVIR setup is depicted in **Figure 1**. A two-slot syringe pump (Gemini 88; KD Scientific, USA) was used to pump the test item and the S/D mixture. Both streams were mixed using a custom-built in-line mixer made of acrylic glass (an example is shown in Figure S1, Supporting Information). A cross-shaped magnet provided active mixing inside the flat cylinder chamber and prototypes with up to four mixing chambers were used. Dynamic mixers were preferred over static mixers because of the low void volume (the main contribution to pre-reactor volume). At the manufacturing scale, the choice will depend on the volumetric flow rate, which dictates CVIR volume. The viscosity of the process intermediate stream and especially the VI chemical(s) would also impact the choice, where viscous streams, such as undiluted S/D chemicals would require a more effective mixer (*i.e.*, longer in the case of static mixers). A static mixer might be used provided that its volume represents a small fraction of the CVIR volume and it can effectively homogenize the process streams. The CVIR consists of a High-Scale 16 column (GE Healthcare Life Sciences, Sweden) packed with $200\text{--}400\,\mu\text{m}$ poly(methyl methacrylate) beads (PLPM-300; Kisker Biotech GmbH, Germany). The beads were packed dry and with the aid of a vibration cage. The resulting bed had a height of 132 mm and a void volume of 10.7 mL. The VI process was operated based on a minimum residence time approach of 60 min for at least 99% of the fluid elements leaving the CVIR, meaning $t_{\min} = t[F_{0.01}] = 60\text{ min}$. Therefore, the volume reactor (V_R) is defined as the volume at which $F_{0.01}$ is reached, which means $V_R = V[F_{0.01}] = 9.40\text{ mL}$. The fact that the $V[F_{0.01}]$ is lower than the void volume results from our goal of minimizing the probability of any given fluid element leaving the CVIR before the minimum incubation time (60 min). Note, however, that a minimum incubation time of 60 min for 100% of

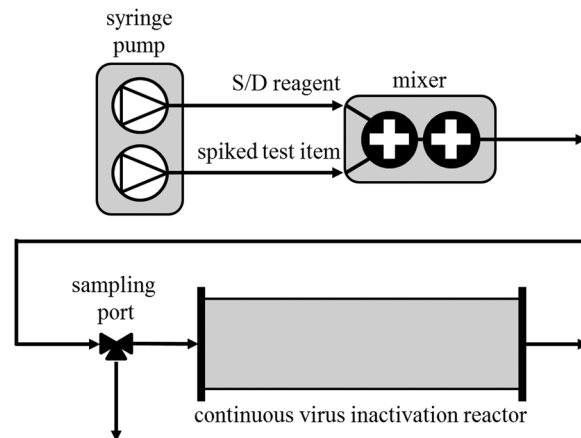


Figure 1. Flow scheme of the setup used for the continuous VI experiments. The setup consists of a double head syringe pump, a two-chamber in-line mixer, a sampling port, and the packed-bed CVIR.

the fluid elements cannot be guaranteed because the tracer concentration falls below the detection limit. Therefore, the maximum claimable logarithmic reduction values (LRV) can be calculated based on the tracer's limit of detection and the assumption that the 1% portion that has incubation times up to 60 min is partially inactivated accordingly. Furthermore, our results show complete inactivation.

S/D Mixing Confirmation: S/D Concentration at the Mixer Outlet Custom-built in-line active mixers with 1–4 chambers were tested. Phosphate-buffer saline (PBS), $10\text{ mM Na}_2\text{HPO}_4\cdot 2\text{H}_2\text{O}$, $1.80\text{ mM KH}_2\text{PO}_4$, 137 mM NaCl , 27 mM KCl , $\text{pH } 6.65 \pm 0.10$ was used as a surrogate of the process intermediate. The S/D stock mixture/PBS volumetric flow ratio was always kept at 1:9, however, flow rates equivalent to 1 min and 60 min incubation were tested. The UV signal at 250 nm was registered at the mixer outlet and the signal fluctuation was used as a proxy for fluctuation of the S/D components' concentrations.

S/D Concentration at the CVIR Outlet A nonspiked test item and the prediluted S/D stock were used to mimic the viral inactivation conditions. A sample was retrieved at the mixer outlet (sampling port in Figure 1) when the CVIR setup was operated first at Q_{total} of 9.40 mL min^{-1} and then at 0.157 mL min^{-1} (corresponding to 1 min and 60 min incubation, respectively). The stream was then redirected through the CVIR and samples were retrieved at the CVIR outlet after 0.5, 1.0, 1.5, 2.0, 3.0, 4.0, and $5.0V_R$ of operation with a Q_{total} of 0.157 mL min^{-1} (corresponding to 60 min). A sample of the nonspiked test item was used as a control. The S/D chemicals' concentrations were determined by high-performance liquid chromatography (HPLC) analysis. The HPLC limits of quantification for TX-100, PS80, and TnBP are $1.0\,\mu\text{g mL}^{-1}$, 0.002% (*w/w*), and $0.5\,\mu\text{g mL}^{-1}$, respectively.

CVIR Characterization: The CVIR was characterized by frontal analysis using a noninteracting tracer (2% (*v/v*) acetone in water). The $t[F_{0.5}]/t[F_{0.01}]$ metric, the column efficiency in

theoretical plate/meter, and the column asymmetry at 10% were determined from the differentiated breakthrough.

CVIR-Induced Virus Loss Control: Control at Extreme Flow Rates A spiked test item and the test item buffer (instead of the prediluted S/D stock) were used to mimic the viral inactivation experiment without S/D chemicals. The CVIR setup was operated at a Q_{total} of $0.157 \text{ mL min}^{-1}$ and 9.40 mL min^{-1} (corresponding to 60 min and 1 min incubation, respectively) with both virus models. Samples for virus titration were collected from the spiked test item, from the mixer outlet stream, from the CVIR outlet stream at $3.0V_R$ of operation (representing steady state), and from the hold control (HC).

Control at a Low Flow Rate A spiked test item and the test item buffer (instead of the pre-diluted S/D stock) were used to replicate the viral inactivation experiment without S/D chemicals. The CVIR setup was operated at Q_{total} of $0.157 \text{ mL min}^{-1}$ (corresponding to 60 min incubation) with both virus models. Samples for virus titration were collected from the spiked test item, from the mixer outlet stream, from the CVIR outlet stream after 1.0, 2.0, 3.0, 4.0, and $5.0V_R$ of operation, and from the HC.

2.2.2. Virus Infectivity Assay

Infectious virus titers were determined by the median tissue culture infective dose assay (TCID_{50}), using eightfold replicates of serial half-log sample dilutions of virus-containing samples that were titrated on the cell lines indicated above. The cells were incubated at 36°C before the cytopathic effect was evaluated by visual inspection under an inverted microscope (Nikon Eclipse TS100; Nikon, Japan). TCID_{50} titers were calculated according to the Poisson distribution and expressed as $\log_{10}(\text{TCID}_{50} \text{ mL}^{-1})$. Virus reduction factors were calculated in accordance with the EU Committee for Proprietary Medicinal Products guidance.^[30]

2.2.3. Viral Inactivation Method

Batch S/D Viral Inactivation: Approximately, 50 mL of the test item was adjusted under stirring to $19 \pm 1^\circ\text{C}$ and then filtered through a $0.2 \mu\text{m}$ syringe filter (Millipore Millex GV PVDF). The determined volume of the filtered test item was spiked at a ratio of 1:30, e.g., 45 mL of the process material were spiked with 1.5 mL of the virus stock suspension. A sample each for virus titration and for the HC was taken (SC was titrated immediately and the HC after the S/D treatment). The weight of the test item (in the closed flask) was determined for calculation of the amount of the S/D stock to be added. The 1/10 pre-diluted S/D stock mixture ($8.39 \pm 1.68 \text{ mg}$) was added per gram of spiked test item, under stirring within 1 min using a syringe, and the actual amount of S/D stock added was determined by back-weighing the syringe. The test item was further incubated under continued stirring at $19 \pm 1^\circ\text{C}$ for $59 \pm 1 \text{ min}$. Samples for TCID_{50} virus titration were drawn

$1\text{--}2 \text{ min}$, $10 \pm 1 \text{ min}$, $30 \pm 1 \text{ min}$, and $59 \pm 1 \text{ min}$ after the addition of S/D stock.

Continuous S/D Viral Inactivation: The CVIR setup was buffered with the test item buffer. The spiked test item and the prediluted S/D stock were loaded into a 100 mL and a 10 mL syringe, respectively. The volumetric flow rates (Q_{total}) were set to 9.40 mL min^{-1} , 0.94 mL min^{-1} , $0.313 \text{ mL min}^{-1}$, and $0.157 \text{ mL min}^{-1}$ to achieve continuous incubation times of 1.0 min, 10 min, 30 min, and 60 min, respectively. Once the custom-built mixer was primed and samples were drawn, the sampling valve was closed and the viral inactivation runs were operated for at least $5V_R$. Samples for virus titration were collected from the spiked test item, from the CVIR outlet stream after 1.0, 2.0, 3.0, 4.0, and $5.0V_R$ of operation and from the HC. Samples containing S/D chemicals were immediately diluted 20-fold to stop the inactivation process. All continuous experiments were performed in a temperature-controlled laminar flow hood where temperatures ranged from 19°C to 22°C . While S/D treatment is effective at temperatures between 2°C and 30°C ,^[13] the temperature chosen would depend on product stability considerations. A Pt-100, temperature probe was attached to the CVIR's wall and the temperature was recorded (μR1000 ; Yokogawa America, USA) throughout the experiment. The average Q_{total} for each V_R was confirmed by periodic gravimetric measurement of the CVIR effluent. The CVIR setup was sanitized between experiments with 0.5 N NaOH for 10 min.

3. Results

3.1. System Controls

3.1.1. Mixing Confirmation

The S/D mixture is viscous and denser than water, mainly due to TX-100 and PS80. The absorbance at 250 nm (A_{250}) was measured at the outlet of a custom-built mixing chamber (Figure S1, Supporting Information) and used as a proxy for the S/D concentration. The A_{250} peak-to-peak amplitude normalized to the average signal (relAP2P) was quantified for different mixers and the extreme flow rates were used (Figure 2A). The mixer prototypes resulted in relAP2P up to 2%, which meant that the S/D chemicals were mixed to $\pm 1\%$ of their target concentration. The two-chamber mixer was selected as a compromise between the robustness of having multiple mixing chambers and the requirement for low prereactor volumes. The total pre-reactor volume including the two-chamber mixer is only 1.09 mL.

The S/D chemicals' concentrations were confirmed independently by HPLC (Figure 2B–D) and normalized by inlet concentration (M1). The S/D chemicals' concentrations were confirmed after the mixer at different flow rates. The concentration shows the start-up phase until $2V_R$ followed by a stage where the S/D concentration is constant at $\pm 10\%$ of their targets. It is noteworthy that the start-up phase of the PS80 concentration is not as evident as for the other VI chemicals. The PS80 outlet concentration rises from 96% (at $0.5V_R$) to 103% of the inlet concentration (at $1.5V_R$). This happens

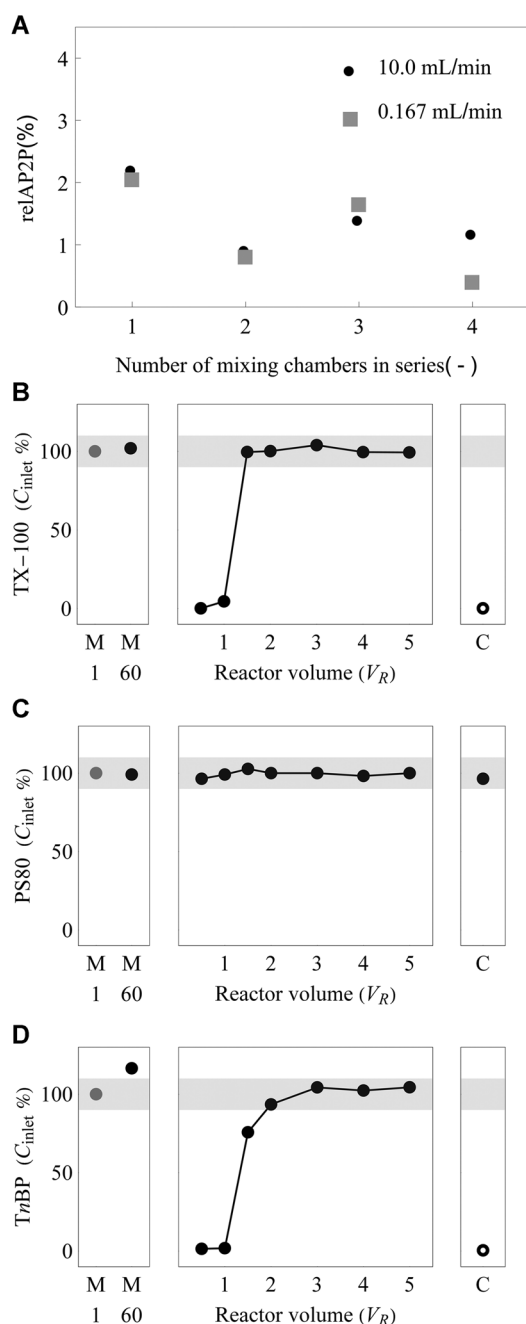


Figure 2. A) Results for the in-line mixing of the S/D mixture with PBS buffer using 1–4 mixing chambers. The relative peak-to-peak amplitude (relAP2P) was calculated from the fluctuation $A_{250\text{ nm}}$ signal for the extreme flow rates used, 0.167 mL min^{-1} and 10.0 mL min^{-1} , equivalent to 60 min and 1 min incubation, respectively. Breakthrough of S/D chemicals. The B) TX-100, C) PS80, and D) TnBP concentrations normalized for their respective inlet concentration. The M1 and M60 denote the samples collected after the mixer/before the CVIR at 9.40 mL min^{-1} and 0.157 mL min^{-1} (gray and black, respectively). The V_R and C denote the samples collected at the outlet of the CVIR and the sample matrix control before the addition of S/D chemicals, respectively. The full points show measured concentration and open points show samples below the LOD. The gray area represents $\pm 10\%$ of the respective inlet concentration (M1).

because all the process-relevant buffers contain 0.1% (v/v) PS80, therefore, the concentration increase upon S/D addition is relatively small ($+0.0138\%$ (v/v)). PS80 is considered a virus-inactivating agent,^[33,34] however, PS80 has much less powerful virus-inactivating potential than TX-100 alone, TnBP alone, or a mixture of TX-100/TnBP.^[33–35] The presence of PS80, especially 0.1% (v/v) PS80, does not lead to significant VI for 60 min incubation time (the longest incubation time studied in the present work). The impact of PS80 present in the process buffers is further discussed in Section 3.1.3. It is worth noting that the mixer outlet sample for TnBP at a low flow rate (sample M60 in Figure 2D) is outside of the $\pm 10\%$ range, however, determination error is likely the cause for the increased concentration. This is supported by the fact that the samples at the CVIR outlet collected during steady state (from 2 until $5V_R$ in Figure 2D) are within the expected range and were generated immediately after M60 at the same flow rates in the same experiment.

3.1.2. CVIR Characterization

The CVIR performance was measured by the $t[F_{0.5}]/t[F_{0.01}]$ metric, column efficiency, and asymmetry (Figure 3). While the column efficiency and asymmetry are well established in column chromatography, $t[F_{0.5}]/t[F_{0.01}]$ is more sensitive for fronting, especially important in cases where minimum residence time is the design criterion. $t[F_{0.5}]/t[F_{0.01}]$ arises from the deviations to the ideal plug flow. $t[F_{0.5}]/t[F_{0.01}]$ is 1.157 at 250 cm h^{-1} , which was used to set the CVIR volume ($V_R = 9.40\text{ mL}$) based on the void volume. This means that the CVIR was operated so that 99% of the stream leaving the reactor at any moment was incubated for 60 min or more. Regardless of the operating linear velocity, the CVIR has an excess of 2000 plate m^{-1} , with increasing efficiency and with lower linear velocities, the asymmetry lies between 1.1 and 1.3, with some variability, which denotes limited tailing.

3.1.3. CVIR-System Controls

The controls without S/D chemicals show that the mixer does not cause any inactivation regardless of the flow rate and the virus model (Figure 4A,B). While product-specific assessment must be carried out, the fact the enveloped virus particles, notably shear-sensitive entities, were not inactivated inside the mixer chamber provides a good indication that the mixer shear might not be harmful to bioactivity of the potential product. At the CVIR outlet, no inactivation was observed for the low incubation time experiments. In contrast, virus loss is observed both after the CVIR and in the HC for the 60 min incubation. This loss is $<1.0\log_{10}$ and, hence, deemed nonsignificant. These titer losses at 60 min incubation were further investigated. An experiment with the same duration and sampling as the VI runs was performed without S/D chemicals (Figure 4C,D). The HCs performed show that the titers for X-MuLV and BVDV decrease over time, the latter being more affected than the former. The comparison between the samples at the CVIR outlet and those from the secondary HC are comparable and well within the expected error for a

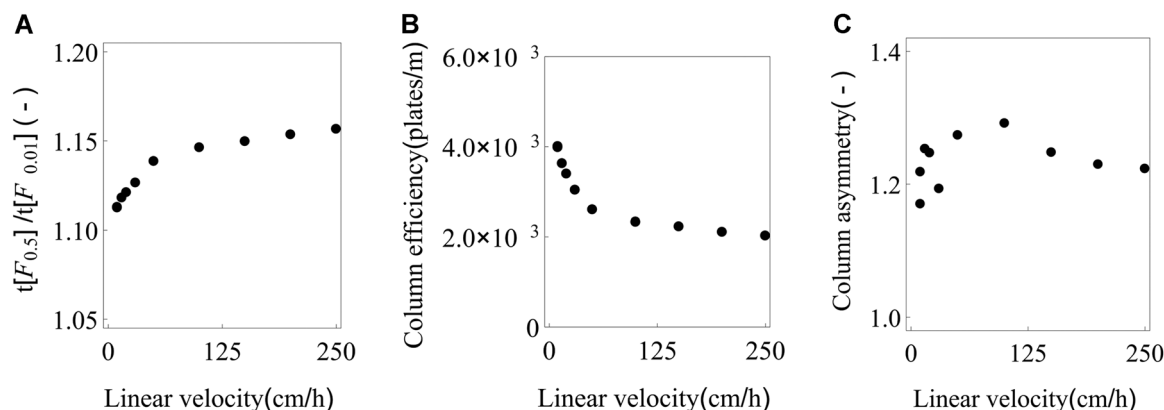


Figure 3. Column characterization by the A) $t[F_{0.5}]/t[F_{0.01}]$ metric, B) column efficiency, and C) peak asymmetry.

titration method.^[31] This suggests that the virus loss might be due to environmental conditions instead of shear-induced damage or adsorption. Most likely the observed BVDV loss ($1.2 \log_{10}$ in the HC) is due to the PS80 present in the process buffers. PS80 can be used without other additional chemicals for VI. The PS80 concentration is usually ten-fold higher, e.g., 1% (v/v), and an incubation time of multiple hours is

required for effective VI.^[15] The fact that the system used for the continuous experiment has no impact on the virus titer is particularly important when claiming comparability with established batch processes. Furthermore, a system that does not contribute to VI ensures that all VI observed before reaching the titration's LOD can be attributed to the S/D action in a process validation study.

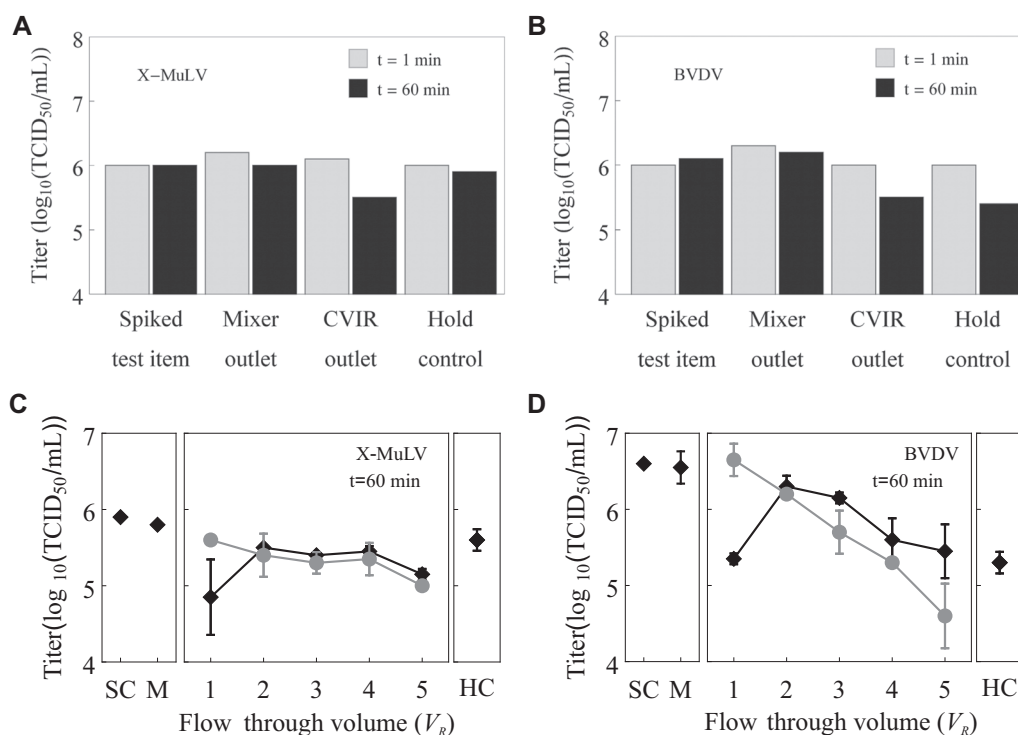


Figure 4. System control runs without S/D chemicals at the extreme incubation times (inversely extreme flow rates) for A) X-MuLV and B) BVDV. The spiked test item, the stream coming from the in-line mixer into the packed-bed CVIR, the stream coming from the packed-bed CVIR after $3V_R$, and the HC were sampled. The runs were performed at two residence times (t_R), 1 min and 60 min (shown in light gray and black, respectively), equal to a flow rate of 9.40 mL min^{-1} and $0.157 \text{ mL min}^{-1}$, respectively. Control experiments without S/D components mimicking the VI experiments for the C) X-MuLV and D) BVDV at the longest incubation time (lowest flow rate). The spiked test item control (SC), the mixer's outlet (M), the continuous reactor's outlet after 1, 2, 3, 4, and $5V_R$ and the HC were sampled. The samples from the reactor outlet (in black) were controlled with a secondary HC (in dark gray) that did not flow through the mixer and reactor but was incubated aside and sampled at the same time as the reactor outlet. The runs were performed at the lowest flow rate ($0.157 \text{ mL min}^{-1}$) meaning a $t_R = 60 \text{ min}$. The runs were performed in duplicate and the error bars represent the standard deviation.

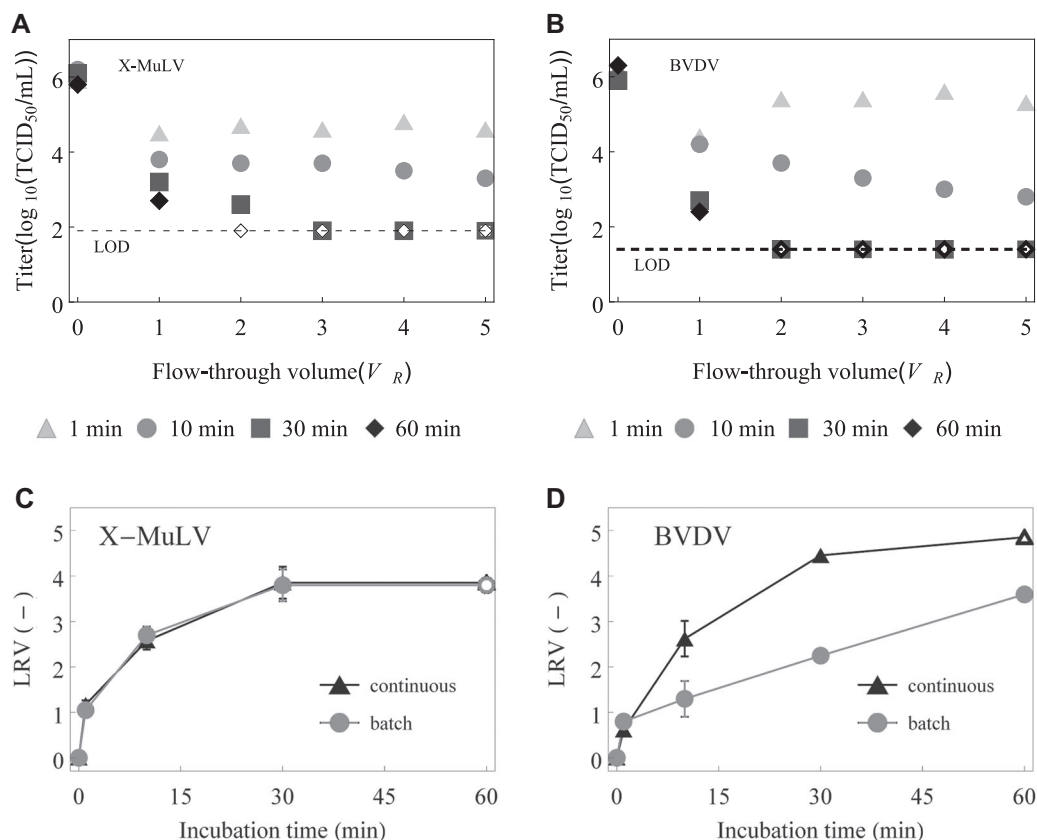


Figure 5. A) X-MuLV and B) BVDV incubation with S/D. Virus titers at the continuous reactor's outlet after 1, 2, 3, 4, and 5 reactor volumes. The sample at $0V_R$ denotes the starting concentration of the spiked test item concentration. The incubation times range from 1 min to 60 min, meaning flow rates from 9.40 mL min^{-1} to $0.157 \text{ mL min}^{-1}$, respectively. The full markers show the virus titer and the open markers represent samples below the LOD. Virus inactivation kinetics with the CVIR compared with a batch process for C) X-MuLV and D) BVDV. The average LRV for the continuous process was calculated considering only samples collected from $2V_R$ onward (steady-state) and is shown in black. The corresponding batch experiment is shown in gray. The full markers show the LRV data points and the open markers represent the minimum LRV due to samples falling below the LOD. The error bars represent the standard deviation ($n = 4$ for continuous and $n = 2$ for batch). The lines link the adjacent points and serve only as visual aids.

3.2. Comparison of Batch and Continuous VI

The virus titer at the CVIR's outlet (Figure 5A,B) shows limited VI for 1 min incubation time. However, with increasing incubation times the virus titers drop below the TCID_{50} LOD, for both model viruses. The BVDV experiment at 10 min incubation time shows titer decrease with increasing operation time, however, the difference between the titer at $2V_R$ and at $5V_R$ is $0.9 \log_{10}$. Therefore, this difference is considered non-significant. The TCID_{50} assay is a cell-based assay and an error of $0.5 \log_{10}$ for a given determination is commonly accepted by the regulatory bodies.^[30] The virus titers at the CVIR outlet is constant after $2V_R$, indicating that the reactor reaches a steady state before or at $2V_R$. It is important to highlight that samples collected at $1V_R$ correspond to the front between the buffer inside the CVIR and the sample mixture with S/D; hence, the S/D chemicals' concentration is below the target, which explains the differences between $1V_R$ and the following V_R points. For this reason, the data at $1V_R$ were excluded from the average titer used to calculate the LRV (Figure 5C,D). Note however that the ramp-up phase before steady state is relatively

fast and it lasts only $2V_R$, which, in the case of 60 min continuous incubation, is 2 h and it would represent only 0.28% out of a 30-day campaign.

The LRV data show time-dependent inactivation kinetics common for VI by S/D treatment^[15,36] and by low-pH treatment.^[37,38] The time-dependent behavior is a key requirement when validating a VI process, and a "biphasic curve" with a fast initial phase and a slower second phase is expected.^[30]

The LRV data generated in continuous operation are compared against data generated in an equivalent batch operation (Figure 5C,D). It should be noted that each of the incubation time points of continuous VI shown in Figure 5C,D result from the respective virus titers obtained during the steady-state phase (from $2V_R$ onward) at the respective incubation time (flow rate). In the case of X-MuLV, the inactivation kinetics obtained for the continuous operation mode matches closely to the one obtained for the batch operation. In the case of BVDV, the inactivation kinetics in continuous mode shows a difference compared with the batch operation mode. Even though both operation modes reach a similar LRV for a 60 min incubation, the inactivation kinetics in

continuous seems to be slightly more efficient for the 10 min and 30 min incubations compared with the batch operation.

It is known that PS80 has limited virus-inactivating potential. However, the PS80-induced inactivation ($0.4 \log_{10}$ after 120 min; see Figure 4C,D) does not explain fully the difference observed between batch and continuous ($2.2 \log_{10}$ after 90 min). Regardless, it is shown that the CVIR is as effective as the batch and both deliver similar LRV after 60 min.

4. Discussion

In this work, an incubation chamber employing a bed of inert nonporous beads was successfully tested for S/D VI using two industry-relevant virus models. The truly continuous nature of the concept was confirmed by steady-state operation after an initial start-up phase. Both the S/D chemicals' concentrations—critical process parameters—at the CVIR's outlet and the virus model titers at the CVIR's outlet are constant from $2V_R$ onward supporting the claim that the CVIR setup reached steady state before $2V_R$. This means a ramp-up phase duration of 120 min if one considers a VI incubation of 60 min. On the basis of product cost considerations and on product dilution/process integration, the CVIR can be prefilled with the desired buffer containing the S/D chemicals at the target concentrations, meaning that the CVIR ramp-up effluent could be further processed instead of being discarded.

An extensive set of control experiments demonstrated that the system itself does not contribute to VI. The data show that neither exposure to shear (in the active mixer) nor virus adsorption inside the column contributes to VI. This means that the entirety of the observed VI can be attributed to S/D treatment in a process validation study, which is a prerequisite for process validation.

Our suggested approach is new and, therefore, questions related with robustness and failure will be raised by both process development scientists and regulatory agencies. Due to the CVIR's similarities with chromatography columns, concerns about packing deterioration during operation, channeling formation, or back pressure increase are understandable. However, one should consider that employed PMMA beads are rigid and non-porous; hence, the chance of bed collapse is lower than when compressible and porous beads are used. Furthermore, CVIR's bed stability is assured by the packing procedure. While under vibration, the top adapter is lowered until physically possible, effectively locking the beads in place. The size of the bead particles used in the study ranges from 200 μm to 400 μm ; such large particle size is also advantageous in case of potential particle formation inside the CVIR, and therefore, reducing the chances of fouling, however, product-specific testing is advised. Because of the bead size, there is reduced pressure drop across the CVIR bed (e.g., 0.583 Pa at 30 cm h^{-1} or $5.34\text{E} + 4 \text{ Pa s m}^{-1}$ for a 13.2 cm bed). Such reduced pressure drop would also allow the implementation of an S/D step earlier in the process to more viscous (or crude) process intermediates. While unlikely, CVIR failure would mean either repacking or replacing the CVIR with a backup unit.

The S/D treatment resulted in inactivation below the titration LOD after 60 min with both virus models despite the fact that the tested S/D chemicals' concentrations are only 5% of those typically used in a GMP manufacturing setting. This emphasizes the robustness of the S/D treatment when manufacturing concentrations are used. Others have shown that the S/D treatment is a fast process and complete inactivation can be achieved within a few minutes.^[13] However, for bioproduction, the S/D treatment is performed for 60 min or more.^[13,35,39] Therefore, to show equivalent efficacy between the established batch process and the new continuous process, we have shown that the VI kinetics in continuous mode is as efficient as in batch mode for comparable incubation times. As discussed before, one could envisage shortening the S/D treatment given the fact that, for historical reasons, the incubation time is longer than needed for full inactivation. However, executing a VI validation study using S/D concentrations typical of manufacturing processes (e.g., 1% (v/v) TX-100 and 0.3% (v/v) TnBP) for short incubations is highly challenging in a batch operation because full inactivation will occur within the first minute.^[13] Here, the CVIR is very advantageous because one can control the incubation time through the flow rate. Therefore, one could perform a VI validation study with incubation times of 15 s, 30 s, 45 s, and 60 s in a precise and reproducible manner and register a fast VI kinetics (a regulatory requirement^[30]) with high S/D chemicals' concentrations.

The LRV obtained in continuous operation mode are comparable with those obtained in batch operation mode for both model viruses. The CVIR is as efficient as the batch process despite the obvious technological differences and advantages. Additionally, packed-bed columns are well-characterized, easy to operate, and common in the biopharma industry, an important factor in technology adoption. The scale of the experiments presented in this work is well suited for process validation studies during process development as an appropriate scale-down model. With this technology now it is possible to put together a fully integrated continuous biomanufacturing train also for acid-labile products.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

bovine viral diarrhea virus, murine leukemia virus, residence time distribution, tri-*n*-butyl phosphate, virus clearance

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